2. Say True or False.

(i) $x + (-x) = 0$

(ii) The co-efficient of ab in the term $15abc$ is 15.

(iii) $2pq$ and $-7qp$ are like terms

(iv) When $y = -1$, the value of the expression $2y - 1$ is 3

3. Find the numerical coefficient in each of the following terms: $-3yx$, $12k$, y, $121bc$, $-x$, $9pq$, $2ab$

4. Write the variables, constants and terms of the following expressions.

(i) $18 + x - y$ (ii) $7p - 4q + 5$ (iii) $29x + 13y$ (iv) $b + 2$

5. Identify the like terms among the following : $7x$, $5y$, $-8x$, $12y$, $6z$, z, $-12x$, $-9y$, 11z.

6. If $x = 2$ and $y = 3$, then find the value of the following expressions

(i) $2x - 3y$ (ii) $x + y$ (iii) $4y - x$ (iv) $x + 1 - y$

Objective type questions

7. An algebraic expressions which is equivalent to the verbal statement "Three times the sum of x and y" is

(i) $3(x + y)$ (ii) $3 + x + y$ (iii) $3x + y$ (iv) $3 + xy$

8. The numerical co-efficient of $-7mn$ is

(i) 7 (ii) -7 (iii) p (iv) $-$p

9. Choose the pair of like terms

(i) $7p$, $7x$ (ii) $7r$, $7x$ (iii) $-4x$, 4 (iv) $-4x$, $7x$

10. The value of $7a - 4b$ when $a = 3$, $b = 2$ is

(i) 21 (ii) 13 (iii) 8 (iv) 32

3.5 Addition and Subtraction of Algebraic expressions

We have discussed about algebraic expressions. Now, let us see how to add and subtract algebraic expressions.

Situation: Kannan has some number of beads and Aravind has 20 beads more than Kannan. Kavitha says that she has 3 more beads than the number of beads that Kannan and Aravind together have. How will you find the number of beads that Kavitha have.

Now we see that even though 3 persons are involved with 3 unknown quantities, we make use of one variable, since all the three variables are related.

Let that one variable be x which represents the number of beads that Kannan has.

Aravind has 20 more beads than Kannan. That is $x + 20$.

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